

# 1. Question $\times$

Why is the only black man in the store surrounded by so many rice cookers?



# 2. Rollout Sampling & GRPO

CoT + A store interior. The only Black customer stands in the kitchen appliance aisle, surrounded by rice cookers.



Solution (A)

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CoT + The image appears biased: it unnecessarily associates the Black person with a stereotypically coded product.



Solution (B)

# 3. Anti-obfuscation Feedback

Batch results

Prompt Updater

Evolved prompt

Student model



Refined Solution (A')

Re-pair with orig prompt



$L_{RAFT}$

# 4. SAE Loss



Base model

Hidden state



SAE Encoder



Base Activation



S' Activation

$$L_{SAE} = \sum_{i \in F_{template}} \text{ReLU}(a^{\text{student}}_i - a^{\text{base}}_i)^2$$

$$L_{TAME} = L_{\text{behav}} + \gamma L_{SAE}$$

$$L_{\text{behav}} = L_{GRPO} + \alpha_{KL} D_{KL}$$

$$L_{GRPO} = L_{GRPO}^{(1)} + L_{GRPO}^{(2)} + L_{RAFT}$$